

## ELECTRON UI

PubMed efetch

**Stage 1**  
Query & Schema  
Definition

QueryBuilder.tsx

**Stage 2**  
Gene Relevance  
Scoring

geneRelevanceScorer.ts

IPC Bridge  
PROGRESS|LOG|RESULT

python-bridge.ts

**Output**  
Results Explorer  
CSV/XLSX Export

Results.jsx

iCite  
PM

**Stage 3**  
Full-Text  
Acquisition

full\_text\_fetcher.py

S2 API  
PMC

**Stage 4**  
PubTator3 NER  
(precision floor)

pubtator3\_tool.py

PubTator3 API

**Stage 5: LLM Extraction**

Stage A: Abstract discovery  
Stage B: Full-text + schema  
+ Figure analysis (Vision)

deterministic seeds gemini\_extractor.py (1,703 LOC)

Gemini Flash API

Orchestrator  
2-4 workers  
600s timeout

orchestrator.py

## VALIDATION

HGNC REST

MyGene.info

**Stage 6a**  
Gene Validation  
HGNC (44,943 local)

gene\_validator.py

**Stage 6b**  
Citation  
Cross-Reference

SequenceMatcher  $\geq 0.85$

**Stage 6c**  
HGVS Variant  
Matching

12+ patterns

**Confidence**  
Gate  $\geq 0.7$

GroundingCheck

**CSV Output**  
Validated, ranked